

LESSON 9.1

SLIDES, FLIPS, AND SPINS



LESSON INTRODUCTION

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

LEARNING TARGETS

- I can show the mapping of a preimage to an image.
- I can describe relationships between figures after a slide, flip, or spin.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

ESSENTIAL QUESTIONS

- Do the side lengths of a shape change in a slide, flip, or spin?
- Do the angle measures of a shape change in a slide, flip, or spin?
- What is the difference between a preimage and an image?
- What are the characteristics that help you identify a transformation as a slide, flip, and spin?

LESSON NARRATIVE

In this lesson, students describe the transformation of a preimage to an image as a slide, flip, or spin. Students discover that the side lengths and angles remain the same when a slide, flip, and spin are performed. Students learn how to map a preimage to an image. In future lessons, they will formally name these movements as a translation, reflection, and rotation.

COMPONENT SUMMARY

9.1.1 WARM-UP (~5 MIN)

Discover what a Nutritionist does

9.1.3 GUIDED INSTRUCTION (10 MIN)

Explore general definitions of preimages, images, and mapping

9.1.5 BRING IT TOGETHER (5 MIN)

Describe relationship between figures after a slide, flip, or spin

9.1.2 COLLABORATION (10 MIN.)

Describe movements as slide, flip, or spin

9.1.4 EXPLORATION (15 MIN)

Measure and compare side lengths and angles of preimages and images

9.1.6 WRAP-UP (~5 MIN)

Reflect on level of interest and understanding

RESOURCES

GLOSSARY

Key Terms:

- Mapping

Additional Terms:

- Preimage
- Congruent
- Image
- Corresponding Points

MATERIALS

- Ruler
- Protractor
- (Optional) Patty Paper
- (Optional) Enlarged Images & Figures on Card Stocks

ADDITIONAL PREPARATION

- Warm up 9.1.1 contains a video to accompany the question prompts

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the movements flip and spin.
- Students may incorrectly assume the image is the preimage.
- Students may say “post-image” instead of just “image.”
- Students may measure the sides or angles incorrectly and think the size of a shape changed.

THINGS TO CONSIDER

- The formal terms: translation, reflection, and rotation will be introduced in Lesson 2. Lesson 1 is about exploring the movements and making connections to prior knowledge.
- Consider providing students with enlarged visual representations of the image in 9.1.2 Collaboration and the figures in 9.1.4 Exploration could be created on card stock and cut out prior to lesson.
- If time is an issue, then consider assigning one of the questions in 9.1.4 Exploration for homework.
- Consider having students use patty paper to copy the images to perform the transformations and/or determine if lengths and angles are congruent.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 9.1.2 Collaboration, ask students to think about and then discuss what movement Felix used. Debrief by having a variety of students share their thinking. Ask students if any movements could be described in more than one way.

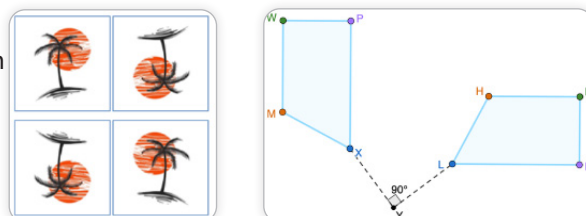
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students are provided opportunities to make connections between the concept of rigid transformations using both art and geometric figures. Students further build their understanding of rigid transformations by mapping the preimage and image. Students begin to build the vocabulary of rigid transformations through drawings.

DIFFERENTIATE INSTRUCTION

Consider providing a visual and kinesthetic entry-point by giving students enlarged card stock images and figures to manipulate in 9.1.2 Collaboration and 9.1.4 Exploration. You could also describe the movements aloud, highlighting key words, to provide an auditory entry-point.



9.1.1 WARM-UP: NUTRITIONIST

QUESTIONING

Consider asking students if they have considered being a nutritionist or if they know a nutritionist.



9.1.1 Warm-Up: Nutritionist



Sasha Bard is a registered dietitian nutritionist who works for the U.S. Department of Agriculture. A nutritionist, or dietitian, is an expert in food, health, and nutrition. Nutritionists are health professionals, so they work in a variety of settings. They work in hospitals helping patients, in grocery stores helping customers, and in gyms to help athletes perform their best.

1. What is something that surprised you about what a nutritionist does?

Answers will vary.

Sasha recommends that students who may be interested in pursuing this field should get out and volunteer at a food pantry or soup kitchen.

2. How do you think volunteering at a food pantry or soup kitchen would be helpful for someone interested in being a nutritionist?

Answers will vary.



9.1.2 Collaboration: Describing Movements

Felix is using palm tree wallpaper squares to create a design on his wall. He experiments with different movements by sliding, flipping, and spinning squares like the one shown.



1. For each movement shown in the table, identify the description of the movement Felix used.

	Movement	Description
A		○ slide ○ flip ○ spin
B		○ slide ○ flip ○ spin
C		○ slide ○ flip ○ spin

2. Compare your responses with your partner and resolve any differences.
3. Choose one of the movements and describe how you determined its type.

I determined movement

○ A

○ B

○ C

 showed a

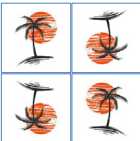
○ slide

○ flip

○ spin

because ...
Answers will vary. The second image was a mirrored version of the first image. I determined movement B showed a slide because the second image was moved over in relation to the position of the first image.

4. Describe which type(s) of movements you think Felix used to create the design shown.



Answers will vary.

9.1.2 COLLABORATION: DESCRIBING MOVEMENTS

TEACHER MOVES

Ask students what they think slide, flip, and spin mean in their own words before engaging in the activity. Consider prompting students to write down their informal personal definitions for each term to reference throughout the lesson and to compare to the definitions of the formal terms developed in Lesson 2. Watch for students who may confuse flips and spins.

DIFFERENTIATE INSTRUCTION

Consider providing visual and kinesthetic entry-points by creating enlarged images of the palm tree (or another image such as the school mascot) on cardstock and prompting students to physically manipulate it.

TEACHER MOVES

Think-Pair-Share

Ask students to think about question 3 and then discuss their answer with their partner. Debrief by having a variety of students share their thinking.

TEACHER MOVES

There are multiple correct answers to this question. The goal is not for students to get this “right,” rather to explore, talk about, and think about different movements that can be made to a figure to change its location and/or orientation.

Students could be given a short time to think/write about how they would describe the movements made to create this and then turn and talk with a partner while the teacher observes and listens to get a sense of what students notice and how they describe the movements. Formal vocabulary around transformations will be introduced in lesson 2. The goal in this lesson is to get students thinking about transformations using language they already know and are familiar/comfortable with using.

9.1.3 GUIDED INSTRUCTION: PREIMAGES, IMAGES, AND MAPPING

ENRICHMENT

- What is the same and what is different between the preimage and the image?
- What movement took place between the preimage and image?

DIFFERENTIATE INSTRUCTION

Describe the movements aloud, highlighting key words, to provide an auditory entry-point.

TEACHER MOVES

To debrief students answers and provide students an opportunity to move during the lesson, ask students to stand up for one option and sit down for the other option. If students are unable to stand up, they can put their thumbs up or down.

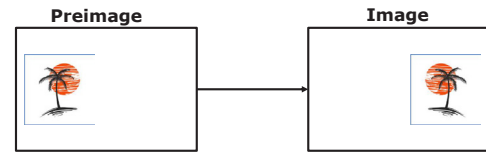
ENRICHMENT

- What is a map?
- How is the definition of a mapping connected to how we use maps in our everyday lives?

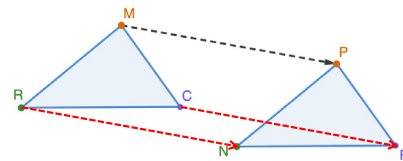


9.1.3 Guided Instruction: Preimages, Images, and Mapping

In geometry, when a figure is transformed from one position to another, the original figure is called the **preimage** and the figure that results after some kind of movement is called the **image**.



Triangle MCR is shown. Consider $\triangle MCR$ the preimage. Imagine a copy of $\triangle MCR$ slides along the dotted ray shown to create a new image, $\triangle PFN$.



1. Discuss with your partner what you notice about the relationship between the preimage and the image. Then, write a brief summary of your discussion.

Answers will vary

Notice that point M on the preimage resulted in point P on the image. These are called **corresponding points**.

2. Complete each statement.

a. Point M on the ☐ image
☒ preimage corresponds to point P on the ☒ image.
☐ preimage.

b. Point F on the ☒ image
☐ preimage corresponds to point ☐ R
☒ C on the ☐ image.
☒ preimage.

c. Point R on the ☐ image
☒ preimage corresponds to point ☐ N
☒ F on the ☒ image.
☐ preimage.

The ray from point M to point P shows the movement of the preimage to the image. This is called a mapping and shows how one point is "mapped" onto another.



A **mapping** is a standardized form of showing how the preimage moves, or transforms, to the image.

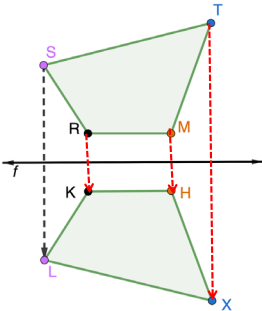
3. Draw rays on the diagram to show the mapping of points C and R to their corresponding points in the image.

The three rays together now show the mapping of the preimage, $\triangle MCR$ to the image, $\triangle PFN$ because all of the vertices are shown to have the same movement.



9.1.4 Exploration: Images and Preimages in Slides, Flips, and Spins

1. Quadrilateral $MRST$ is flipped across line f to create quadrilateral $HKLX$ so that point S aligns with point L .



- a. The dotted ray shows the mapping of point S to point L . Complete the mapping of the vertices of quadrilateral $MRST$ to the vertices of quadrilateral $HKLX$.

- b. Complete the table by working with your partner to find each measure using rulers and protractors.

Side Length	Measure (cm)	Side Length	Measure (cm)
Segment MR	1.95 cm	Segment HK	1.95 cm
Segment RS	1.9 cm	Segment KL	1.9 cm
Segment ST	4 cm	Segment LX	4 cm
Segment TM	2.75 cm	Segment XH	2.75 cm
Angle	Measure (°)	Angle	Measure (°)
$\angle MRS$	123°	$\angle HKL$	123°
$\angle RST$	72°	$\angle KLX$	72°
$\angle STM$	58°	$\angle LXH$	58°
$\angle TMR$	107°	$\angle XHK$	107°

- c. What do you notice about the side lengths and angle measures of quadrilateral $MRST$ and quadrilateral $HKLX$?
They are the same.

9.1.4 EXPLORATION: IMAGES AND PREIMAGES IN SLIDES, FLIPS, AND SPINS

TEACHER MOVES

The teacher could structure this as a partner activity or as a gallery walk. If a partner activity, each pair of partners would complete the entire exploration. As a gallery walk, partners or groups would complete one of the three, and then the teacher would structure a gallery walk for students to see what other groups explored and make note of the same information in their workbook before the 9.1.5 Bring It Together.

DIFFERENTIATE INSTRUCTION

Consider providing visual and kinesthetic entry-points by creating enlarged figures on cardstock and prompting students to physically manipulate it.

DIFFERENTIATE INSTRUCTION

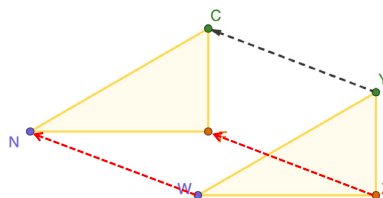
Consider reviewing how to use a ruler and a protractor. The main point of this table is to show students that the lengths and angles in the preimage and image are the same. Consider prompting students to round their measurements to the nearest tenth or whole number.

9.1.4 EXPLORATION: IMAGES AND PREIMAGES IN SLIDES, FLIPS, AND SPINS (CONTINUED)

ENRICHMENT

Why do the side length and angle measurements not change when we slide the figure?

2. The image, $\triangle NCF$, is shown after sliding the preimage, $\triangle WYZ$, along the dotted ray.



- a. Complete the mapping from $\triangle WYZ$ to $\triangle NCF$.
- b. Complete the table by working with your partner to find each measure using rulers and protractors.

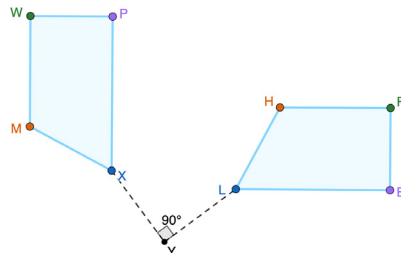
Side Length	Measure (cm)	Side Length	Measure (cm)
Segment WY	4.8 cm	Segment NC	4.8 cm
Segment YZ	2.4 cm	Segment CF	2.4 cm
Segment WZ	4.15 cm	Segment NF	4.15 cm
Angle	Measure (°)	Angle	Measure (°)
$\angle WYZ$	60°	$\angle NCF$	60°
$\angle YZW$	90°	$\angle CFN$	90°
$\angle ZWY$	30°	$\angle FNC$	30°

- c. Discuss the relationship between the side lengths and angle measures of $\triangle WYZ$ and $\triangle NCF$ with your partner.

QUESTIONING

As an extension, have students reflect on real-world examples of slides, flips, and spins that they see in their everyday lives.

3. Trapezoid $LHFB$ is spun around point Y to create trapezoid $XMWP$.



The angle shown, 90° , provides a mapping of one set of vertices for the spin. Each point on trapezoid $LHFB$ forms a 90° with the corresponding point on trapezoid $XMWP$ with point Y as its vertex.

Complete the table by working with your partner to find each measure using rulers and protractors.

Side Length	Measure (cm)	Side Length	Measure (cm)
Segment LH	2.2 cm	Segment XM	2.2 cm
Segment HF	2.6 cm	Segment MW	2.6 cm
Segment FB	1.9 cm	Segment WP	1.9 cm
Segment BL	3.6 cm	Segment PX	3.6 cm
Angle	Measure (°)	Angle	Measure (°)
$\angle BLH$	62°	$\angle PXM$	62°
$\angle LHF$	118°	$\angle XMW$	118°
$\angle HFB$	90°	$\angle MWP$	90°
$\angle FBL$	90°	$\angle WPX$	90°



9.1.5 Bring It Together: Describing the Relationship Between Figures After a Slide, Flip, or Spin

Review the three movements explored in today’s lesson – slides, flips, and spins.

1. Identify whether each movement preserves distance (side length) and angle measure. Complete the table below by selecting “yes” or “no” based on your observations.

Movement	Is distance (side length) preserved?	Are angle measures preserved?
Slide	☐ Yes ☐ No	☐ Yes ☐ No
Flip	☐ Yes ☐ No	☐ Yes ☐ No
Spin	☐ Yes ☐ No	☐ Yes ☐ No

Figures are **congruent** if all corresponding parts of the figures are congruent.

2. Complete the statements.
When a slide, flip, or spin is performed on a figure, the

corresponding side lengths of the preimage and image

☐ are
☐ are not the same length and the angle measures

☐ are
☐ are not the same measure. Therefore, the

preimage and image are ☐ congruent
☐ similar figures.

9.1.5 BRING IT TOGETHER: DESCRIBING THE RELATIONSHIP BETWEEN FIGURES AFTER A SLIDE, FLIP, OR SPIN

ENRICHMENT

- What do you notice about your answers?
- What conclusions can you make about the preimages and images of slides, flips, and spins?

TEACHER MOVES

Ask students to copy these statements in writing and read them aloud. Repeated writing and reading of the statements will reinforce and solidify student understanding of the concepts.

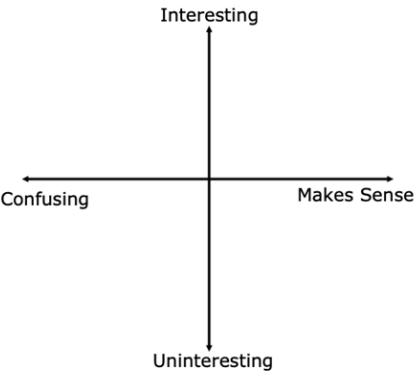


9.1.6 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and how much sense you made of today’s learning targets.

Answers will vary.

- ☐ I can show the mapping of a preimage to an image.
- ☐ I can describe relationships between figures after a slide, flip, or spin.



9.1.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider prompting students to plot three separate points for slide, flip, and spin. They can label the points or use a different color or symbol for each.